

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Name: T. Ganesh (192425246)

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent.

(5 Marks)

Answer:

A steady increase in reward from 10 to 45 across training episodes is a classic learning curve signature, and it directly reflects the agent's policy improving over time.

- **Interpretation of the trend:** the monotonic (gradual, non-erratic) rise shows the agent is progressively learning a better mapping from states to actions — early episodes reflect near-random exploratory behaviour (low reward), while later episodes reflect exploitation of a refined, learned policy (high reward).
- **Link to policy improvement:** under algorithms such as Q-learning or policy gradients, each update nudges the policy toward actions with higher estimated value; a rising reward curve is empirical evidence that this iterative policy-

improvement step is working as intended (consistent with the Generalized Policy Iteration framework — evaluate, then improve, repeat).

- **Caveat:** the trend alone does not prove the policy is optimal — it only shows improvement is occurring; convergence must be confirmed by checking that the reward plateaus (stops increasing) over further episodes.
- **Conclusion:** The increasing reward trend indicates healthy, effective learning and successful policy improvement, but the agent should be trained further and monitored for a plateau before concluding the policy has reached its optimum.

2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**

Answer:

The results directly illustrate the exploration–exploitation trade-off that governs ϵ -greedy action selection.

- **Why $\epsilon=0.1$ performs best:** a low ϵ means the agent exploits its current best-known action ~90% of the time and explores randomly only ~10% of the time; once the agent has learned reasonably accurate value estimates, this ratio lets it capitalize on good actions most of the time while still occasionally sampling alternatives, maximizing cumulative reward.
- **Why $\epsilon=0.5$ performs worst:** with $\epsilon=0.5$, the agent chooses a random (often suboptimal) action half the time, so even a well-learned policy is 'sabotaged' by excessive random exploration — cumulative reward suffers because too much reward-generating potential is sacrificed for exploration that is no longer needed once the environment is reasonably well understood.
- **General principle:** exploration is essential early in training (to discover good actions) but should be reduced over time (often via ϵ -decay) once the agent has enough information, since continued high exploration after learning has converged only wastes reward.
- **Conclusion:** $\epsilon = 0.1$ provides the better performance in this scenario because it keeps enough exploitation of the learned policy while retaining minimal, sufficient exploration; $\epsilon = 0.5$ over-explores and unnecessarily reduces cumulative reward.

3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the performance difference between these algorithms and determine which algorithm converges faster based on the observed results. **(5 Marks)**

Answer:

Q-Learning and SARSA are both temporal-difference control algorithms, but they differ in how they estimate the value of the next action, which explains the observed gap in reward.

- **Q-Learning (off-policy):** updates its Q-values using the maximum Q-value of the next state, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot \max_{a'} Q(s',a') - Q(s,a)]$, effectively learning the value of the greedy (optimal) policy regardless of the exploratory actions actually taken — this tends to push value estimates (and hence achieved reward) toward the optimal policy faster.
- **SARSA (on-policy):** updates using the actual next action chosen by the current (exploratory) policy, $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \cdot Q(s',a') - Q(s,a)]$, so its value estimates are influenced by the exploration noise (e.g., ϵ -greedy randomness) built into the behaviour policy, making it slightly more conservative and typically yielding marginally lower reward, especially near risky states.
- **Convergence speed:** because Q-Learning directly targets the optimal action-value function irrespective of the exploration strategy, it generally converges to a higher-reward policy faster than SARSA across increasing episode counts (100 $\rightarrow$ 200 $\rightarrow$ 300), consistent with the observed pattern of Q-Learning consistently outperforming SARSA.
- **Conclusion:** Based on the consistently higher rewards at each checkpoint, Q-Learning converges faster than SARSA in this setting, though SARSA's on-policy, more conservative updates can be preferable in environments where exploratory actions carry real risk (e.g., safety-critical tasks).

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

Answer:

The discount factor γ determines how much weight the agent gives to future rewards relative to immediate ones, and the observed trend (reward rising with γ) reflects this directly.

- **$\gamma = 0.2$ (low):** the agent is highly myopic, focusing almost entirely on immediate reward and largely ignoring the long-term consequences of its actions; this produces a lower average reward whenever the task genuinely requires multi-step planning to reach high-value outcomes.
- **$\gamma = 0.5$ (moderate):** the agent balances immediate and future rewards but still discounts distant rewards fairly heavily, so it captures more long-term value than $\gamma=0.2$ but still under-weights delayed payoffs relative to a truly far-sighted policy.
- **$\gamma = 0.9$ (high):** the agent values future rewards almost as much as immediate ones, encouraging it to learn strategies that may sacrifice small short-term rewards for much larger long-term returns — this generally yields the highest average reward when the environment's true optimal policy depends on long-horizon planning.
- **Conclusion:** $\gamma = 0.9$ is the most suitable discount factor here because it allows the agent to properly account for long-term reward optimization, matching the

observed trend of higher average reward with higher γ ; however, γ too close to 1 in noisy/continuing tasks can slow convergence, so the exact best value should still be tuned per environment.

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

Answer:

A success rate rising from 45% at 50 episodes to 92% at 200 episodes is a strong, substantial improvement that reflects effective policy learning, though it should be interpreted carefully before declaring the policy fully optimal.

- **Magnitude of improvement:** a jump of 47 percentage points over 150 additional episodes indicates the agent has learned to reliably solve the task far more often than early in training, confirming that its policy updates are producing genuinely better decisions, not just noise.
- **Is the policy optimal?:** 92% success is very good but not 100%, so the policy is likely near-optimal but may still fail in certain edge-case states, under insufficient exploration of rare situations, or due to inherent environment stochasticity that caps achievable success below 100%.
- **Supporting evidence needed:** to confirm convergence to an optimal (or near-optimal) policy, the success-rate trend should be checked for a plateau beyond 200 episodes — if it keeps climbing, the policy hasn't fully converged; if it stabilizes around 90–95%, that plateau itself is evidence of convergence to a strong (possibly optimal, given environment noise) policy.
- **Conclusion:** The data shows clear evidence of substantial learning and policy improvement; the agent has very likely learned a near-optimal policy, but full optimality should only be confirmed once the success rate plateaus over further episodes rather than continuing to rise.

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

Answer:

The learning rate α controls the size of each update step to the agent's value/policy estimates, and its effect on stability in CartPole follows a classic 'too low / just right / too high' pattern.

- **$\alpha = 0.01$ (too low):** updates are very small, so the agent learns cautiously and stably but slowly — it may take far more episodes to converge, and within a fixed training budget it may not yet have reached a strong, stable policy, showing lower apparent stability simply due to under-training.

- **$\alpha = 0.05$ (balanced):** this learning rate is large enough to let the agent adapt its policy quickly toward better balancing strategies, yet small enough to avoid overshooting good parameter values — this combination is why it achieves the highest observed stability, i.e., a good trade-off between learning speed and steady convergence.
- **$\alpha = 0.1$ (too high):** large updates can overshoot good solutions and cause oscillation in the learned value/policy estimates — the pole-balancing policy can swing between good and poor performance from episode to episode, reducing stability even though early learning may look fast.
- **Conclusion:** $\alpha = 0.05$ offers the best trade-off between convergence speed and stability: it is fast enough to learn an effective CartPole policy within a reasonable number of episodes while avoiding the destabilizing oscillations seen at the higher learning rate.

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario.

(5 Marks)

Answer:

The three exploration strategies differ in how they balance trying new actions against exploiting the currently best-known action, and ϵ -greedy's structured randomness explains its superior cumulative reward here.

- **Random exploration:** chooses actions with no regard to their estimated value at all, so the agent wastes a large fraction of its experience on clearly poor actions throughout training, resulting in the lowest cumulative reward of the three.
- **Softmax exploration:** chooses actions probabilistically in proportion to their estimated Q-values (via a Boltzmann distribution), which is more informed than random exploration, but if the temperature parameter is not perfectly tuned it can either over-explore (too flat a distribution) or under-explore (too peaked), making its performance sensitive to that hyperparameter.
- **ϵ -greedy exploration:** exploits the best known action with high probability ($1-\epsilon$) while reserving a small, fixed probability ϵ for pure exploration; this simple, robust rule captures most of the value of the currently best policy while still gathering enough exploratory data to keep improving, and is less sensitive to mistuning than softmax's temperature parameter.
- **Conclusion:** ϵ -greedy achieves the best cumulative reward because it reliably exploits the best known action most of the time while retaining just enough controlled exploration, giving a more consistent balance than the unstructured randomness of random exploration or the tuning-sensitive softmax approach.

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy.

(5 Marks)

Answer:

A reduction from 50 steps to 15 steps to reach the goal after 300 episodes is strong behavioural evidence of policy improvement, since fewer steps directly means a more efficient (and typically closer-to-optimal, in a shortest-path sense) route through the maze.

- **Interpretation:** at the start, the agent is essentially exploring randomly and taking a long, inefficient path (or wandering) to reach the goal; as training progresses, the agent's Q-values (or policy parameters) are refined so it consistently selects actions that lead more directly to the goal, shrinking the path length.
- **Relation to convergence:** if 15 steps closely matches (or equals) the theoretical shortest path length in the maze, this is strong evidence the agent has converged to at or near the optimal policy; if the true shortest path is significantly less than 15 steps, the agent has improved substantially but has not yet fully converged to the optimal solution.
- **Further verification:** checking whether the step count continues to decrease, stays flat at 15, or fluctuates over additional episodes would confirm whether 15 steps represents a converged, stable optimal policy or an intermediate plateau.
- **Conclusion:** The large reduction in steps clearly indicates significant policy improvement and likely convergence toward an efficient, near-optimal policy, but this should be confirmed by comparing 15 steps against the maze's known shortest path and checking that performance remains stable in later episodes.

9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective.

(5 Marks)

Answer:

The discount factor γ in SARSA plays the same fundamental role as in any RL algorithm — it determines how strongly future rewards influence the current value estimate — and the result (highest long-term reward at $\gamma=0.95$) reflects that this task rewards far-sighted behaviour.

- **Role of γ :** γ close to 0 makes the agent prioritize immediate reward almost exclusively, ignoring the long-term consequences of its actions, while γ close to 1 makes the agent weigh distant future rewards nearly as heavily as immediate ones, encouraging strategies that optimize for the entire trajectory rather than the next step alone.
- **Comparing 0.4, 0.7, 0.95:** $\gamma=0.4$ is strongly myopic and likely misses reward-maximizing strategies that require patience or delayed payoff; $\gamma=0.7$ is a moderate middle ground; $\gamma=0.95$ lets the agent properly account for rewards many steps into the future, which is why it captures the greatest long-term return when the task's true reward structure depends on long-horizon planning.
- **Why 0.95 works well specifically for SARSA:** since SARSA is on-policy and updates using the actual next action taken, a higher γ helps it more fully credit the value of the exploratory-but-ultimately-successful trajectories it experiences, better aligning its learned values with the true long-term returns of its behaviour policy.

- **Conclusion:** $\gamma = 0.95$ is the most effective discount factor here because the task rewards patience and long-term planning; it allows the SARSA agent to properly value delayed rewards rather than being dominated by short-term gains, matching the observed highest long-term reward.

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning.

(5 Marks)

Answer:

A reward pattern that fluctuates early on but stabilizes after 500 episodes in a stochastic environment is a well-known and reassuring signature of successful, robust policy learning.

- **Why early fluctuation occurs:** at the start of training, the agent is both exploring (trying varied, sometimes poor actions) and learning from a noisy, stochastic environment where the same action can yield different outcomes; both factors combine to produce highly variable episode rewards early on.
- **Why stabilization matters:** once the agent's policy and value estimates converge, its action choices become consistent and well-adapted to the environment's underlying dynamics, so reward variance shrinks even though the environment itself remains stochastic — the agent has effectively learned to handle the randomness (e.g., by choosing actions that are robust to noisy outcomes) rather than being thrown off by it.
- **Robustness interpretation:** stability after 500 episodes despite ongoing environmental stochasticity indicates the learned policy generalizes well across the range of random outcomes the environment produces, rather than having merely memorized one lucky trajectory — this is a hallmark of a robust, well-converged policy.
- **Conclusion:** The transition from fluctuating to stable rewards is strong evidence of policy improvement and convergence: the agent has learned a policy that consistently performs well despite the environment's inherent randomness, demonstrating both learning success and robustness.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand